Responses generated by the LLM after the analysis of call paths and its corresponding system tree from cube profiles.

"Path" corresponds to the path of the files as uploaded on sciebo
"Time taken" corresponds to the full time taken from hitting the analyze button till finishing analysis

CALL TREE WITH A SINGLE ROOT

1. **Path: 1/CIAO-IO/scorep-ciao-pavel/mini-lpt-480 → profile.cubex**
   Time taken: ~2 mins
   1 root; 68 call paths, 480 system tree elements

Analyzing call tree...
====================================
REGION 1
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler"
severity: 5.2%
slowest_ranks: [21,41,42,53,54,70,73,74,85,86,90]
reason: A small subset of ranks (around 10) take noticeably longer than the bulk, indicating a load-imbalance where those ranks hold more work.
investigate: compare the amount of mesh elements or particle counts assigned to ranks 21,41,42,53,54,70,73,74,85,86,90 with the average per-rank workload.
fix: repartition the domain so that the heavy mesh/particle blocks currently on the identified slow ranks are split and redistributed to under-utilized ranks.

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REGION 2
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/MPI_File_set_view"
severity: 7.5%
slowest_ranks: [160]
reason: One rank (160) runs noticeably longer than the rest, indicating an I/O load imbalance for that MPI process.
investigate: compare the amount of data each rank accesses and the size of the MPI_File_set_view for rank 160 versus the other ranks using the I/O profiling counters.
fix: redistribute the file view offsets so that each rank reads a more equal portion of the file, balancing the I/O workload across ranks.

====================================
REGION 3
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/scp_lpt_step/MPI_File_write_all"
severity: 2.5%
slowest_ranks: [160]

reason: The runtimes are tightly clustered around 250 ms with only rank 160 exceeding 257 ms, indicating a modest load imbalance.
investigate: compare per-rank iteration counts and data volumes for rank 160 against the average ranks to see why it spends more time.
fix: redistribute a portion of the mesh or I/O workload from rank 160 to neighboring ranks to equalize computation time.

Finished analyzing all selected regions.

2. **Path: 1/CIAO-IO/scorep-ciao-pavel/mini-lpt-720 → profile.cubex**
   Time taken: ~1 min
   1 root; 68 call paths, 720 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler"
severity: 5.86%
slowest_ranks: [420, 451, 489, 520, 545]
reason: A small set of ranks run noticeably longer than the bulk, producing a right-skewed tail in the runtime distribution.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks against the average to see if they receive more work or larger messages.
fix: redistribute the computational work (e.g., mesh partitions or particle sets) from the identified slow ranks to the lighter ranks to even out the load.

====================================
REGION 2
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/MPI_File_set_view"
severity: 4.5%
slowest_ranks: [57, 108, 172]
reason: most ranks run in a narrow band around 150 µs while a few ranks exceed 158 µs, indicating a mild load imbalance.
investigate: compare the per-rank file view offsets and amount of data each rank reads/writes for the listed slowest ranks versus the average ranks.
fix: repartition the MPI-IO file views so that the data volume assigned to the slowest ranks is reduced and balanced across all ranks.

====================================
REGION 3
----

call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/scp_lpt_step/MPI_File_write_all"
severity: 2.1%
slowest_ranks: [350]
reason: The runtimes are tightly clustered around 420 ms with only a small tail exceeding 430 ms, indicating a mild load imbalance.
investigate: compare the per-rank I/O volumes and collective write timings for rank 350 against the bulk of the ranks to see why it takes longer.
fix: redistribute the file write offsets so that the large I/O chunk handled by rank 350 is split more evenly among neighboring ranks.

Finished analyzing all selected regions.

3. **Path: 1/CIAO-IO/scorep-ciao-pavel/mini-lpt-960 → profile.cubex**
   Time taken: ~3 mins
   1 root; 68 call paths, 960 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler"
severity: 7.0%
slowest_ranks: [945]
reason: Most ranks cluster around 14.5–14.9 s with a single outlier near 15.76 s, indicating a modest load imbalance.
investigate: compare per-rank iteration counts and workload sizes for rank 945 against the median ranks to confirm it performed more work.
fix: redistribute the mesh or data items assigned to rank 945 so its workload matches that of the typical rank.

====================================
REGION 2
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/MPI_File_set_view"
severity: 32.6%
slowest_ranks: [216]
reason: The runtime distribution shows many ranks clustered around 70–78 s while a few ranks (including rank 0) have near-zero time, indicating significant load imbalance.
investigate: compare per-rank iteration counts and workload sizes for rank 216 versus the lower-runtime ranks to see why rank 216 takes substantially longer.
fix: rebalance the data or computation assigned to rank 216 (e.g., redistribute mesh partitions) so its workload matches that of the lighter-loaded ranks.

=====================================
REGION 3
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/scp_lpt_step/MPI_File_write_all"
severity: 1.2%
slowest_ranks: [224,215,209]
reason: A small subset of ranks consistently take noticeably longer than the average, indicating an imbalance in the I/O workload.
investigate: compare the amount of data written and the number of collective I/O calls performed by the listed ranks against the faster ranks.
fix: redistribute the file view or chunk assignments so that the amount of data each MPI rank writes is more evenly balanced.

Finished analyzing all selected regions.

4. **Path: 1/CIAO-IO/scorep-ciao-pavel/mini-lpt-1200-12.10.10 → profile.cubex**
   Time taken: ~3 mins
   1 root; 68 call paths, 1,200 system tree elements

Analyzing call tree...

=====================================
REGION 1
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler"
severity: 3.6%
slowest_ranks: [245]
reason: The runtime distribution shows most ranks clustered near the mean (~14.8 s) while a few ranks, especially rank 245, run noticeably longer, indicating a moderate load imbalance.
investigate: compare per-rank iteration counts and the amount of computational work assigned to rank 245 against the surrounding ranks to verify the extra load.
fix: rebalance the data (e.g., mesh or particle partitions) so that rank 245 receives less work and its runtime aligns with the average.

=====================================
REGION 2
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/MPI_File_set_view"
severity: 9.6%
slowest_ranks:
[40,60,70,80,90,100,110,120,150,170,180,190,200,210,220,230,240,250,260,270,280,290,300,310,320,330,340,350,360,370,380,390,400,410,420,430,440,450,460,470,480,490,500,510]
reason: The runtimes are clustered around 160-170 s with a long tail of ranks exceeding 179 s, indicating a moderate imbalance where a minority of ranks take noticeably longer.

investigate: compare the per-rank runtime values (especially the ranks listed above) with their assigned iteration counts or workload sizes to see why those ranks run longer.
fix: redistribute the computational work (e.g., mesh cells or particle batches) from the identified slow ranks to the faster ones so that all ranks have a more even workload.

====================================
REGION 3
----
call_path: "Invisible call tree root/arts_scorep_advance_iter_lpt_nocompiler/scp_lpt_step/MPI_File_write_all"
severity: 3.5%
slowest_ranks: [210,70,80,100,120]
reason: The majority of ranks execute in a narrow band (~540–570 µs) while a few ranks take noticeably longer and two ranks are idle, indicating load imbalance.
investigate: compare per-rank I/O operation counts, data volume written and file offsets for the ranks whose runtimes exceed 565 µs.
fix: redistribute the MPI-File write chunks so that the slowest ranks receive smaller file sections, balancing the I/O workload across all ranks.

Finished analyzing all selected regions.

5. **Path: 1/hybrid-POP/comparison/pop_compare/CIAO/scorep_JUWELS_384x1_144577_trace_WITH_Y2_OPTION → profile.cubex**
   Time taken: ~5 mins
   1 root; 3,281 call paths, 384 system tree elements

NOTE: A large call tree- 3281 regions. Throws an "Error handling/couldn't process request" error. Hit analyze button 3-4 times and it should work. No idea why this happens.

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/MAIN__"
severity: 99%
slowest_ranks: [59,66,73,63,92,67]
reason: Most ranks finish in ~6e-5 s while a few ranks take >0.02 s, indicating a severe load imbalance.
investigate: compare per-rank iteration counts and workload sizes for ranks 59, 66, 73, 63, 92 and 67 to verify why they execute far more work than the others.
fix: redistribute the heavy mesh/particle partitions from the overloaded ranks to achieve a more even workload distribution.
====================================
REGION 2

----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
arts_func.arts_scalar_step_finite_/scalar_weno5_func.scalar_weno5_residual_/
scalar_weno5_func.scalar_weno5_coeff_"
severity: 3.30%
slowest_ranks: [96]
reason: One MPI rank (96) runs significantly longer than the average, indicating a workload imbalance.
investigate: compare per-rank computational work (e.g., number of cells or particles processed) and iteration counts for rank 96 against the other ranks.
fix: repartition the mesh/particles owned by rank 96 and distribute them to neighboring ranks to balance the workload.

======================================
REGION 3
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
arts_func.arts_scalar_step_finite_/scalar_weno5_func.scalar_weno5_inverse_"
severity: 5.7%
slowest_ranks: [48, 96, 144, 192, 206, 336, 383]
reason: The runtime distribution shows a few ranks taking up to ~6% longer than the average, indicating load imbalance.
investigate: compare per-rank iteration counts and computational work for the listed ranks against the median ranks to see why they run slower.
fix: redistribute the mesh or computational cells assigned to the slowest ranks so that each rank handles a more even amount of work.

======================================
REGION 4
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
combustion_step_finite_/finitechem.finitechem_step_/finitechem.finitechem_cvode_solve_"
severity: 14.2%
slowest_ranks: 9, 24, 180
reason: most ranks cluster around 200 s while a few exceed 240 s, indicating a moderate load imbalance.
investigate: compare per-rank element/particle counts and local work estimates for ranks 9, 24, and 180 against the average to verify they carry disproportionate work.
fix: repartition the mesh or particle ownership so that the heavy ranks transfer part of their domain to under-utilized ranks, equalising the per-rank runtime.
======================================
REGION 5

----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
vdf_velocity.velocity_pressure_/vdf_pressure.pressure_solve_/
hypre_amg_func.hypre_amg_solve_"
severity: 9.6%
slowest_ranks: [12,42,43,47,31]
reason: The runtime distribution shows a moderate imbalance with a handful of ranks taking noticeably longer than the average.
investigate: compare per-rank iteration counts and mesh partition sizes for ranks 12, 31, 42, 43, and 47 against the overall average to verify they are overloaded.
fix: repartition the mesh so that the computational load on ranks 12, 31, 42, 43, and 47 is reduced and distributed more evenly across all MPI ranks.

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REGION 6
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
vdf_velocity.velocity_pressure_/vdf_pressure.pressure_solve_/
hypre_amg_func.hypre_amg_solve_/MPI_Waitall"
severity: 7.8%
slowest_ranks: [23,24,27,31,53,76,99,124,147,172,197,221,244,267,291]
reason: most ranks cluster around 410-440 ms while a few exceed 455 ms, indicating a noticeable tail of slower MPI-Waitall calls.
investigate: compare the number of pending non-blocking communications and the size of message buffers on the listed slow ranks versus the median ranks.
fix: redistribute pending MPI requests or adjust the domain decomposition so that the ranks with the longest wait times handle fewer or smaller messages.

Finished analyzing all selected regions.

6. **Path: 1/hybrid-POP/comparison/pop_compare/CIAO/scorep_JUWELS_384x1_145407_profile_WITHOUT_Y2_OPTION → profile.cubex**
   Time taken: ~5 mins
   1 root; 3,265 call paths, 384 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/MAIN__"
severity: 98.4%

slowest_ranks: [29]
reason: A single rank (29) consumes orders of magnitude more time than the vast majority, indicating a severe load imbalance.
investigate: compare the per-rank computational workload (e.g., mesh cells or particle counts) assigned to rank 29 with that of the low-time ranks to verify the imbalance.
fix: redistribute the data (e.g., mesh partitions) so that the heavy work currently on rank 29 is shared more evenly across all ranks.

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REGION 2
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
arts_func.arts_scalar_step_finite_/scalar_weno5_func.scalar_weno5_residual_/
scalar_weno5_func.scalar_weno5_coeff_"
severity: 4.9%
slowest_ranks: [192]
reason: most ranks cluster around 50 s, but rank 192 runs noticeably longer, indicating an isolated workload imbalance.
investigate: compare the per-rank iteration counts and mesh partition sizes for rank 192 against its neighboring ranks to see why it incurs extra work.
fix: rebalance the mesh partition by moving a portion of rank 192’s cells to adjacent ranks, equalizing the computational load.

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REGION 3
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
combustion_step_finite_/finitechem.finitechem_step_/finitechem.finitechem_cvode_solve_"
severity: 58%
slowest_ranks: [222,175,127,16,126]
reason: The runtime distribution shows a few ranks taking more than double the average while many ranks finish in a few seconds, indicating a strong load imbalance.
investigate: compare per-rank computational workload (e.g., mesh cells or particle counts) for ranks 222, 175, 127, 16, and 126 against the low-runtime ranks.
fix: redistribute the mesh/particle partitions so that the heavy partitions assigned to those ranks are spread more evenly across all MPI ranks.

====================================
REGION 4
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/

combustion_step_finite_/finitechem.finitechem_step_/parallel_m.parallel_sum_real_0d_/
parallel_m.parallel_allreduce_real_0d_/MPI_Allreduce"
severity: 8.3%
slowest_ranks: 289,49,54,291,290,294,52,50,193
reason: A subset of ranks spends ≈10 % more time than the average, indicating a noticeable but not extreme load imbalance.
investigate: compare per-rank iteration counts and workload sizes of the listed slowest ranks against the median ranks to see why they take longer.
fix: redistribute mesh partitions or particle ownership from the identified slowest ranks to give them a lighter workload.

==
REGION 5
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
vdf_velocity.velocity_pressure_/vdf_pressure.pressure_solve_/
hypre_amg_func.hypre_amg_solve_"
severity: 13.2%
slowest_ranks: 96
reason: The runtime distribution shows a few ranks (e.g., rank 96) taking significantly longer (~480 s) while most ranks finish around 410 s, indicating load imbalance.
investigate: compare the number of AMG matrix rows or work assigned to rank 96 with that of the median ranks using the profiling data.
fix: rebalance the AMG matrix partition to give rank 96 a smaller portion of the linear system, reducing its compute load.

==
REGION 6
----
call_path: "Invisible call tree
root/MAIN__/arts_IP_simulation_run_/arts_simulation.arts_advance_iter_/
vdf_velocity.velocity_pressure_/vdf_pressure.pressure_solve_/
hypre_amg_func.hypre_amg_solve_/MPI_Waitall"
severity: 4.3%
slowest_ranks: [290]
reason: most ranks take ~500 ms while rank 290 takes ~531 ms, indicating a modest outlier that creates a slight load imbalance.
investigate: compare the number of AMG iterations and local matrix sizes on rank 290 with those on typical ranks to see why it spends ~20 ms longer.
fix: redistribute a portion of the AMG matrix rows from rank 290 to neighboring ranks to equalize the solve workload.

Finished analyzing all selected regions.

7. **Path: 1/hybrid-POP/comparison/pop_compare/ICON/scorep_icon_24p192_trace → profile.cubex**
   Time taken: ~3 mins
   1 root; 4,373 call paths, 192 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/MAIN__"
severity: 36.13%
slowest_ranks: [0,48,49,50,54]
reason: The distribution is dominated by a single rank whose runtime is far higher than the rest, indicating a severe load imbalance.
investigate: compare per-rank iteration counts and workload sizes for the identified slow ranks (especially rank 0) against typical ranks to reveal the excess computation.
fix: redistribute the work assigned to rank 0 (e.g., split its mesh or data partition) so that its workload matches the average across ranks.

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REGION 2
----
call_path: "Invisible call tree root/MAIN__/MPI_Init"
severity: 0.2%
slowest_ranks: [180,181,182,183,184,185]
reason: The per-rank runtimes are almost identical, indicating the MPI_Init call is uniformly fast across ranks with only a tiny deviation.
investigate: compare the MPI_Init entry-exit timestamps for all ranks, focusing on the few ranks whose timings are at the high end to verify they are not delayed by OS or network latency.
fix: ensure that every rank reaches MPI_Init simultaneously (e.g., by inserting a barrier before MPI_Init) to eliminate the negligible skew.

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REGION 3
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_nh_stepping.perform_dyn_substepping_/mo_solve_nonhydro.solve_nh_"
severity: 3.4%
slowest_ranks: [45, 131, 120, 167, 70]
reason: The runtime distribution shows one rank with essentially no work and a few ranks taking noticeably longer than the bulk, indicating load imbalance across MPI ranks.

investigate: compare the number of grid cells or physical elements assigned to each rank, focusing on ranks 45, 131, 120, 167, and 70 versus the idle rank 0.
fix: repartition the atmospheric mesh so that the high-runtime ranks receive fewer cells and the idle rank receives a share of the work, achieving a more even per-rank workload.

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REGION 4
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call_path: "Invisible call tree
root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/
mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/
mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/
mo_echam_phy_bcs.echam_phy_bcs_global_/mo_bc_aeropt_kinne.read_bc_aeropt_kinne_/
mo_bc_aeropt_kinne.read_months_bc_aeropt_kinne_/
mo_read_interface.read_dist_real_3d_time_/mo_read_interface.read_dist_real_3d_extdim_/
mo_read_netcdf_distributed.distrib_read_real_4d_/
mo_read_netcdf_distributed.distrib_read_real_4d_multi_var_/
mo_communication.exchange_data_r3d_/MPI_Waitall"
severity: 4.36%
slowest_ranks: 131
reason: The runtime distribution shows a moderate imbalance where the slowest rank (131) executes about 4 % longer than the average, while several ranks are noticeably faster (~2.6-2.9 s versus ~3.6-3.8 s).
investigate: compare the size of the atmospheric mesh or number of grid columns assigned to rank 131 with those assigned to the faster ranks to see if the domain decomposition is uneven.
fix: rebalance the domain decomposition so that rank 131 receives fewer grid columns, equalising computational load across MPI ranks.

Finished analyzing all selected regions.

8. **Path: 1/hybrid-POP/KKR → kkr_16x4-1_profile.cubex**
   Time taken: ~2 mins
   1 root; 561 call paths, 64 system tree elements

   Analyzing call tree...

==
REGION 1
----
call_path: "Invisible call tree root/kkr.x"
severity: 75.6%
slowest_ranks: [0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60]
reason: The runtime is concentrated on a few ranks while most ranks report zero time, indicating extreme load imbalance.

investigate: compare per-rank iteration counts or assigned mesh cells between the listed slow ranks and the idle ranks.
fix: repartition the mesh or workload so that all MPI ranks receive a roughly equal number of cells/iterations.

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REGION 2
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1a.main1a_/mod_tmatnewsolver.tmat_newsolver_/mod_rllsll.rllsll_/!$omp parallel/!$omp barrier/!$omp task"
severity: 1.6%
slowest_ranks: [48, 49, 50, 51, 52]
reason: The runtimes are tightly clustered with only about a 1.6% slowdown on the slowest ranks, indicating a mild imbalance.
investigate: compare per-rank workload sizes or iteration counts for ranks 48-52 to identify why they take slightly longer.
fix: redistribute a small portion of the computational domain from the slowest ranks to faster ones to equalize their runtimes.

==
REGION 3
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1b.main1b_/mod_kloopz1.kloopz1_qdos_/mod_kkrmat01.kkrmat01_/mod_dlke0.dlke0_/mod_dlke1.dlke1_"
severity: 76.4%
slowest_ranks: [0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60]
reason: A few MPI ranks consume the entire runtime while the majority perform no work, indicating severe load imbalance.
investigate: compare per-rank iteration counts and the amount of computational work assigned to the listed ranks versus the idle ranks.
fix: redistribute the computational domain (e.g., mesh or k-point slices) so that the heavy ranks receive fewer elements and idle ranks are given work.

==
REGION 4
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1b.main1b_/mod_kloopz1.kloopz1_qdos_/mod_kkrmat01.kkrmat01_/mod_inversion.inversion_"
severity: 33.6%
slowest_ranks: [0,4,20,48,52,56]
reason: The runtime distribution shows a small group of ranks taking roughly 60-63 s while the majority finish in ~34-38 s, indicating load imbalance.

investigate: compare the amount of k-point or matrix work assigned to the listed slow ranks against that of typical ranks using the profiling counters.
fix: rebalance the k-point or matrix partitions by moving portions of the heavy workload from the overloaded ranks to the underloaded ranks.

====================================
REGION 5
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1c.main1c_/mod_rhovalnew.rhovalnew_/mod_rllsll.rllsll_/!$omp parallel/!$omp barrier/!$omp task"
severity: 31.4%
slowest_ranks: [0,4,16,20,32,36,48,52,56,60]
reason: A minority of ranks take roughly 30 % longer than the average, showing a clear load-imbalance where those ranks execute significantly more work.
investigate: compare per-rank iteration counts and computational workload (e.g., mesh cells or particle numbers) of the listed ranks against the faster ranks using the profiling data.
fix: redistribute the mesh or data partitions so that the heavy ranks receive fewer elements, balancing the workload across all MPI ranks.

====================================
REGION 6
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1c.main1c_/mod_rhovalnew.rhovalnew_/mod_rhooutnew.rhooutnew_/mod_intcheb_cell.intcheb_cell_"
severity: 41.4%
slowest_ranks: 49,53,1,57,5
reason: The runtimes form two distinct groups, with a subset of ranks taking roughly twice as long as the majority, indicating a clear load imbalance.
investigate: compare per-rank iteration counts and the number of mesh cells or particles assigned to ranks 49, 53, 1, 57, and 5 versus the lower-runtime ranks.
fix: redistribute mesh partitions or particle ownership so that the high-runtime ranks receive fewer elements, balancing computational work across all MPI ranks.

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REGION 7
----
call_path: "Invisible call tree root/kkr.x/MAIN__/mod_main1c.main1c_/mod_rhovalnew.rhovalnew_/mod_rhooutnew.rhooutnew_/mod_intcheb_cell.intcheb_cell_/mod_intcheb_cell.intcheb_complex_"
severity: 47.5%
slowest_ranks: 48

reason: A small subset of ranks (e.g., rank 48) spend over 105 s while the majority spend around 40 s, indicating a strong load imbalance.
investigate: compare the per-rank workload sizes (e.g., number of mesh cells or particles) for rank 48 against the typical ranks to identify the excess work.
fix: redistribute mesh partitions or particle ownership from rank 48 to less-loaded ranks to achieve a more even runtime distribution.

Finished analyzing all selected regions.

9. **1/hybrid-POP/KKR/case 2 → profile.cubex**
   Time taken: ~4 mins
   1 root; 69 call paths, 64 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/kkr.x"
severity: 79%
slowest_ranks: [0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60]
reason: The runtime distribution is extremely skewed, with a handful of ranks performing all the computation while the majority remain idle.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks to verify that they carry significantly more work than the zero-runtime ranks.
fix: redistribute the mesh partitions more evenly across MPI ranks to balance the computational load.

====================================
REGION 2
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1a/!$omp parallel/!$omp barrier/!$omp task"
severity: 7.5%
slowest_ranks: 9-16
reason: The runtime distribution shows a group of ranks (9-16) executing noticeably longer than the others, indicating load imbalance.
investigate: compare per-rank iteration counts and workload sizes for ranks 9-16 versus the faster ranks to confirm uneven work distribution.
fix: rebalance the domain or loop partitioning to assign fewer elements to ranks 9-16, spreading the workload more evenly across all MPI ranks.

====================================
REGION 3

----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1b"
severity: 65.75%
slowest_ranks: 0,4,8,12,16,20,24,28,32,36,40,44,48,52,56
reason: A few ranks spend about 30-33 s while the remaining ranks spend only ~4.5 s, indicating a severe load imbalance.
investigate: compare per-rank iteration counts and the amount of computational work assigned to the ranks listed as slowest.
fix: redistribute the heavy mesh or domain partitions from the slowest ranks to achieve a more even workload across all MPI ranks.

====================================
REGION 4
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1b/invmod_0/zgetrf"
severity: 46.0%
slowest_ranks: [0,4,8,12,16,20,24,28,32,36,40,44,48,52,56,60]
reason: The runtimes exhibit a bimodal distribution with a group of ranks taking ~45-50 s while the rest take ~20 s, indicating strong load imbalance.
investigate: compare per-rank computational workload or data partition sizes for the listed high-runtime ranks against the low-runtime ranks.
fix: redistribute the heavy mesh or particle partitions from the slow ranks to achieve a more even workload across all MPI ranks.

====================================
REGION 5
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1b/invmod_0/zgetrs/!$omp parallel"
severity: 37.0%
slowest_ranks: [0]
reason: The runtimes form two distinct groups, with many ranks around 31–33 s and a few near 60 s, showing a strong load imbalance.
investigate: compare per-rank iteration counts and workload sizes for rank 0 versus the other ranks to identify the excessive work.
fix: rebalance the domain decomposition to distribute the work of rank 0 more evenly across all MPI ranks.

====================================
REGION 6
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1c"
severity: 40.0%
slowest_ranks: [4,8,12,16,20,24,28,32,36,40,44,48,52,56,60]
reason: The runtimes form two distinct groups, with many ranks around 36-38 s and a subset around 70-74 s, indicating a clear load imbalance.

investigate: compare per-rank iteration counts and workload sizes for the listed ranks versus the faster ranks to see why they take twice as long.
fix: repartition the domain or mesh so that the computational work currently assigned to the slowest ranks is divided more evenly among all MPI ranks.

==
REGION 7
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1c/!$omp parallel/!$omp barrier/!$omp task"
severity: 36.0%
slowest_ranks: [0,40,44,48,32,52,16,36,24]
reason: A small subset of ranks take roughly 30 seconds longer than the average, while most ranks cluster around 45–46 seconds, indicating significant load imbalance.
investigate: compare the amount of computational work (e.g., grid cells or particle counts) assigned to ranks 0,40,44,48,32,52,16,36,24 with that of the typical rank.
fix: repartition the mesh/particle ownership so that the heavy ranks receive fewer cells/particles and the lighter ranks receive more, equalizing their runtimes.

==
REGION 8
----
call_path: "Invisible call tree root/kkr.x/main/main_scf/main1c/!$omp parallel/!$omp barrier/!$omp task"
severity: 36.0%
slowest_ranks: [0, 32, 40, 44, 48]
reason: A few ranks (especially rank 0) run far longer than the bulk of ranks, showing a clear load-imbalance.
investigate: compare the number of OpenMP tasks or mesh/particle elements assigned to the slowest ranks versus the typical ranks.
fix: repartition the mesh or redistribute particle ownership so that the heavy ranks receive fewer elements and the workload is balanced across all MPI ranks.

Finished analyzing all selected regions.

10. Path: 1/ hybrid-POP /nekbone.hyb/scorep_nekbone_1p32x64_sum → profile.cubex
Time taken: ~4 mins
1 root; 222 call paths, 2048 system tree elements

Analyzing call tree...

==
REGION 1
----
call_path: "Invisible call tree root/nekbone"

severity: 57%
slowest_ranks: 0
reason: One MPI rank (rank 0) takes roughly twice the time of the typical rank, while most ranks record near-zero time, indicating a severe load imbalance.
investigate: compare the per-rank call counts and execution times for this call between rank 0 and the other ranks to confirm that rank 0 does significantly more work.
fix: redistribute the portion of the mesh or computation handled by rank 0 so that the work is spread more evenly across all ranks.

====================================
REGION 2
----
call_path: "Invisible call tree root/nekbone/cg/!$omp parallel @cg.input.prep.f:46/glsc3i/!$omp critical @omp.input.prep.f:49"
severity: 29.3%
slowest_ranks: [1275]
reason: most ranks cluster around 0.5-0.7 s, but one rank takes ~0.85 s, indicating a clear load imbalance.
investigate: compare per-rank execution times and the amount of mesh elements or particles assigned to rank 1275 versus the median ranks.
fix: repartition the mesh/particle data so that rank 1275 receives a smaller share, bringing its runtime closer to the average.

====================================
REGION 3
----
call_path: "Invisible call tree root/nekbone/cg/!$omp parallel @cg.input.prep.f:46/glsc3i/!$omp barrier @omp.input.prep.f:53"
severity: 0.6%
slowest_ranks: [1050,1051,1052]
reason: most ranks run at ~1.229 s while a few exceed 1.236 s, indicating a modest tail where a handful of ranks take noticeably longer.
investigate: compare per-rank computation and communication volumes for ranks 1050-1052 against the median ranks to see what extra work or message traffic they incur.
fix: move a portion of the mesh elements owned by ranks 1050-1052 to neighboring ranks so that each rank handles a more equal number of elements.

====================================
REGION 4
----
call_path: "Invisible call tree root/nekbone/cg/!$omp parallel @cg.input.prep.f:46/axi"
severity: 1.2%
slowest_ranks: [26]
reason: most ranks finish within a narrow band (~2.53-2.55 s) while rank 26 runs modestly longer (~2.57 s), indicating a slight workload imbalance.
investigate: compare the number of elements or mesh cells assigned to rank 26 with those of the typical ranks to see if it owns more work.
fix: split the mesh partition assigned to rank 26 and redistribute a portion of its cells to neighboring ranks to equalize the per-rank workload.

====================================

REGION 5

----

call_path: "Invisible call tree root/nekbone/cg/!$omp parallel @cg.input.prep.f:46/axi/gs_op/gs_aux/pw_exec/!$omp barrier @gs.c:638"
severity: 3.8%
slowest_ranks: [1000,1001,1002,1003,1004]
reason: a small subset of ranks run noticeably longer than the average, indicating a modest load imbalance
investigate: compare per-rank iteration counts and workload sizes for the listed ranks to see why they take longer
fix: redistribute mesh partitions more evenly across MPI ranks to equalize the work per rank

Finished analyzing all selected regions.

11. **Path: 1/ICON/MPI+OMP/scorep-20171020_1139_7954530684728313 → profile.cubex**
    Time taken: ~2 mins
    1 root; 8,287 call paths, 36 system tree elements

Analyzing call tree...

====================================

REGION 1

----

call_path: "Invisible call tree root/MAIN__"
severity: 95.8%
slowest_ranks: 0,6,12,18,24,30
reason: The runtime is dominated by a few ranks while most ranks are idle, indicating a severe load imbalance.
investigate: compare per-rank iteration counts or workload sizes for ranks 0, 6, 12, 18, 24, 30 against the near-zero runtimes of the other ranks.
fix: repartition the data or loop iterations to distribute the computational work more evenly across all MPI ranks.

====================================

REGION 2

----

call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_nh_stepping.perform_dyn_substepping_/mo_solve_nonhydro.solve_nh_/!$omp parallel @mo_solve_nonhydro.f90:2125/!$omp do @mo_solve_nonhydro.f90:2151"
severity: 1.11%
slowest_ranks: [18, 30]
reason: The runtimes are tightly clustered around the mean with only a ~1 % deviation for the slowest ranks, indicating a mild imbalance.

investigate: compare the mesh or column counts and solver iteration numbers assigned to ranks 18 and 30 with those of the faster ranks.
fix: repartition the atmospheric mesh so that the work (e.g., number of columns or grid cells) assigned to ranks 18 and 30 is reduced and redistributed to achieve a more even load.

====================================
REGION 3
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp parallel @mo_interface_iconam_echam.f90:518/!$omp do @mo_interface_iconam_echam.f90:519/mo_echam_phy_main.echam_phy_main_/mo_psrad_radiation.psrad_radiation_/mo_psrad_interface.psrad_interface_/mo_psrad_lrtm_driver.lrtm_driver_/mo_psrad_lrtm_solver.lrtm_solver_"
severity: 1.32%
slowest_ranks: [32]
reason: The runtimes differ by only ~0.18 s, showing a mild imbalance with rank 32 slightly slower than the rest.
investigate: compare per-rank computational workload and iteration counts for rank 32 versus other ranks to identify the extra work causing its longer runtime.
fix: redistribute the mesh or data chunks assigned to rank 32 to neighboring ranks to achieve a more even load distribution.

====================================
REGION 4
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp parallel @mo_interface_iconam_echam.f90:518/!$omp do @mo_interface_iconam_echam.f90:519/mo_echam_phy_main.echam_phy_main_/mo_psrad_radiation.psrad_radiation_/mo_psrad_interface.psrad_interface_/mo_psrad_srtm_driver.srtm_"
severity: 0.98%
slowest_ranks: [33]
reason: The runtimes vary by less than 1 % across ranks, indicating a largely balanced workload.
investigate: compare per-rank computational work (e.g., number of grid cells or particle counts) and iteration counts for rank 33 versus the other ranks.
fix: repartition the mesh or data domain to shift a small portion of rank 33’s work onto neighboring ranks.

Finished analyzing all selected regions.

12. Path: 1/ICON/OMP/scorep_icon_Ox18_sum → profile.cubex
Time taken: ~2 mins
1 root; 6,744 call paths, 18 elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/MAIN__"
severity: 94.44%
slowest_ranks: [0]
reason: The runtime is almost entirely spent on rank 0 while all other ranks report zero time, indicating extreme load imbalance.
investigate: compare per-rank iteration counts and computational work assigned to rank 0 versus the idle ranks to confirm the workload disparity.
fix: redistribute the work of rank 0 (e.g., split its mesh or loop iterations) so that multiple MPI ranks share the load evenly.

====================================
REGION 2
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_nh_stepping.perform_dyn_substepping_/mo_solve_nonhydro.solve_nh_/!$omp parallel @mo_solve_nonhydro.f90:2125/!$omp do @mo_solve_nonhydro.f90:2151"
severity: 1.58%
slowest_ranks: 0
reason: Rank 0 takes noticeably longer than all other ranks, indicating an outlier workload causing imbalance.
investigate: compare per-rank iteration counts and mesh/column workload sizes for rank 0 versus the other ranks to see why it executes more work.
fix: repartition the atmospheric grid so that the portion assigned to rank 0 is split among neighboring ranks, achieving a more even distribution of computational work.

====================================
REGION 3
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp

parallel @mo_interface_iconam_echam.f90:518/!$omp do
@mo_interface_iconam_echam.f90:519/mo_echam_phy_main.echam_phy_main_/
mo_psrad_radiation.psrad_radiation_/mo_psrad_interface.psrad_interface_/
mo_psrad_lrtm_driver.lrtm_driver_/mo_psrad_lrtm_solver.lrtm_solver_"
severity: 0.67%
slowest_ranks: [11]
reason: The runtimes are tightly clustered with only about a 0.7 % deviation from the average, showing little imbalance across MPI ranks.
investigate: compare per-rank iteration counts and workload sizes for rank 11 against the other ranks to verify that it is doing slightly more work.
fix: shift a small portion of the mesh or computational tasks from rank 11 to neighboring ranks to equalize the workload.

====================================
REGION 4
----
call_path: "Invisible call tree
root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/
mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/
mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp
parallel @mo_interface_iconam_echam.f90:518/!$omp do
@mo_interface_iconam_echam.f90:519/mo_echam_phy_main.echam_phy_main_/
mo_psrad_radiation.psrad_radiation_/mo_psrad_interface.psrad_interface_/
mo_psrad_srtm_driver.srtm_"
severity: 0.67%
slowest_ranks: [0]
reason: The runtimes are tightly clustered (standard deviation ≈0.09 s) with only a ~0.7 % slower rank, indicating a minor load imbalance.
investigate: compare the number of grid cells or computational work assigned to rank 0 versus the other ranks using the domain-decomposition profiling data.
fix: shift a small portion of the grid cells from rank 0 to neighboring ranks to equalize the workload.

Finished analyzing all selected regions.

13. **Path: 1/ICON/OMP+HWC/scorep_icon_Ox9_sum+HWC → profile.cubex**
    Time taken: ~ 2 mins
    1 root; 5,625 call paths, 9 system tree elements

Analyzing call tree...
====================================
REGION 1
----
call_path: "Invisible call tree root/MAIN__"
severity: 89%
slowest_ranks: [0]
reason: Rank 0 does all the work while the other ranks are idle, indicating severe load imbalance.

investigate: compare per-rank iteration counts and workload sizes to verify why rank 0 has much higher runtime than the others.
fix: redistribute the workload (e.g., mesh partitions or data blocks) more evenly across MPI ranks.

====================================
REGION 2
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp parallel @mo_interface_iconam_echam.f90:518/!$omp do @mo_interface_iconam_echam.f90:519"
severity: 0.31%
slowest_ranks: [0]
reason: The runtimes are nearly identical across ranks, showing a well-balanced execution with only ~0.3 % deviation.
investigate: compare per-rank iteration counts and workload sizes for rank 0 against the other ranks to confirm the tiny imbalance.
fix: retain the current domain decomposition since it already provides balanced runtimes.

====================================
REGION 3
----
call_path: "Invisible call tree root/MAIN__/mo_atmo_model.atmo_model_/mo_atmo_nonhydrostatic.atmo_nonhydrostatic_/mo_nh_stepping.perform_nh_stepping_/mo_nh_stepping.perform_nh_timeloop_/mo_nh_stepping.integrate_nh_/mo_interface_iconam_echam.interface_iconam_echam_/!$omp parallel @mo_interface_iconam_echam.f90:518/!$omp do @mo_interface_iconam_echam.f90:519/mo_psrad_radiation.psrad_radiation_/mo_psrad_interface.psrad_interface_"
severity: 0.08%
slowest_ranks: [6]
reason: The runtimes across ranks differ by only about 0.09 s, showing an almost uniform workload with negligible imbalance.
investigate: compare per-rank iteration counts and computational work for rank 6 against the other ranks to verify the slight runtime difference.
fix: shift a small fraction of mesh cells or computational tasks from rank 6 to neighboring ranks to equalize the runtime.

Finished analyzing all selected regions.

14. Path: 1/Karolina/scorep_clover_leaf_8_sum → profile.cubex
Time taken: ~ 2 mins
1 root; 1,311 call paths, 16 system tree elements

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/clover_leaf"
severity: 80.7%
slowest_ranks: [10]
reason: The runtime is dominated by a single rank while many ranks have near-zero work, indicating a severe load imbalance.
investigate: compare per-rank iteration counts and computational work assigned to rank 10 versus the idle ranks to verify the imbalance.
fix: redistribute the mesh or domain partition so that the workload handled by rank 10 is shared among multiple ranks.

====================================
REGION 2
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/build_field_"
severity: 51%
slowest_ranks: [8,9,10,11,12,13,14,15]
reason: Half of the ranks have zero runtime while the other half each consume ~43 s, showing a severe load imbalance.
investigate: compare per-rank iteration counts and mesh element assignments between the idle ranks (0-7) and the busy ranks (8-15).
fix: redistribute the mesh partitions more evenly across all MPI ranks to equalize computational load.

====================================
REGION 3
----
call_path: "Invisible call tree root/clover_leaf/KERNELS/COMPUTE IDLE"
severity: 6.56%
slowest_ranks: [0,1,2,3]
reason: The first four ranks run noticeably longer than the others, showing a clear load imbalance.
investigate: compare the amount of computational work or mesh cells assigned to ranks 0-3 versus ranks 4-15 to verify the imbalance.
fix: redistribute mesh partitions from ranks 0-3 to the underutilized ranks to balance the workload.

Finished analyzing all selected regions.

15. Path: 1/Karolina/scorep_clover_leaf_16_sum → profile.cubex
Time taken: ~4 mins
1 root; 1,818 call paths, 32 system tree elements

Analyzing call tree...

======================================
REGION 1
----
call_path: "Invisible call tree root/clover_leaf"
severity: 83.1%
slowest_ranks: [9]
reason: The runtime is concentrated on a single rank while the majority of ranks spend virtually no time, indicating severe load imbalance.
investigate: compare the number of leaf-cell iterations and the amount of data processed on rank 9 versus the idle ranks to confirm the disproportionate work distribution.
fix: redistribute the leaf cell partitions more evenly across MPI ranks so that rank 9 shares its workload with neighboring ranks.

======================================
REGION 2
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/clover_module_clover_barrier_/MPI_Barrier"
severity: 74.69%
slowest_ranks: [8,9,10,11,12,13,14,15]
reason: The runtimes are highly skewed, with half the ranks idle (0 s) while the others spend about 70 s, indicating a severe load imbalance.
investigate: compare per-rank computation work or data partitions for ranks 8-15 versus the idle ranks to verify that the heavy ranks carry disproportionate workload.
fix: redistribute the computational domain or data partitions to spread work more evenly across all MPI ranks, reducing the load on ranks 8-15.

======================================
REGION 3
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/build_field_"
severity: 50.1%
slowest_ranks: 8,9,10,11,12,13,14,15,24,25,26,27,28,29,30,31
reason: half of the ranks record zero runtime while the other half spend the full ~79 s, indicating an extreme load imbalance.
investigate: compare per-rank iteration counts or field block sizes between the listed slow ranks and the idle ranks to confirm that work is unevenly partitioned.
fix: repartition the field mesh so that work is spread evenly across all MPI ranks, moving portions from the overloaded ranks to the previously idle ranks.

=====================================
REGION 4
----
call_path: "Invisible call tree root/clover_leaf/KERNELS/COMPUTE IDLE"
severity: 3.4%
slowest_ranks: [0,1,2,3,16,17,18,19]
reason: The runtimes split into a higher-time group (~84 s) and a lower-time group (~79 s), showing a clear load imbalance.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks against the faster ranks.
fix: redistribute mesh partitions or computational domains from the slowest ranks to achieve a more even work distribution.

Finished analyzing all selected regions.

16. **Path: 1/Karolina/scorep_clover_leaf_32_sum → profile.cubex**
    Time taken: ~3 mins
    1 root; 1,311 call paths, 64 system tree elements

Analyzing call tree...

=====================================
REGION 1
----
call_path: "Invisible call tree root/clover_leaf"
severity: 84.5%
slowest_ranks: 24,47,41,40,58
reason: A few ranks take disproportionately longer than the average, indicating strong load imbalance.
investigate: compare the amount of work (e.g., mesh cells or particles) assigned to ranks 24,47,41,40,58 with that of the other ranks.
fix: rebalance the mesh partitions to distribute work more evenly across all MPI ranks.

=====================================
REGION 2
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/clover_module_clover_barrier_/MPI_Barrier"
severity: 62.19%
slowest_ranks: 8,9,10,11,12,13,14,15,40,41,42,43,44,45,46,47,56,57,58,59,60,61,62,63
reason: The runtimes show a clear bimodal distribution with many ranks at zero time and a subset around 17 s, indicating severe load imbalance.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks to determine why they incur ~17 s while many ranks have no work.

fix: rebalance the domain decomposition to distribute the work of ranks 8-15, 40-47, and 56-63 evenly across all MPI ranks.

======================================
REGION 3
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/build_field_"
severity: 50.2%
slowest_ranks: 11,9,10,8
reason: The runtime distribution shows half the ranks idle (0 s) while the active ranks take about 22 s, indicating severe load imbalance.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks
fix: redistribute the computational domain (e.g., mesh partitions) to give the idle ranks comparable work and balance the ~22 s runtime across all ranks

======================================
REGION 4
----
call_path: "Invisible call tree root/clover_leaf/KERNELS/COMPUTE IDLE"
severity: 10.0%
slowest_ranks: 1,17,18,19,20
reason: A small group of ranks runs about 10 % longer than the average, revealing a clear load-imbalance.
investigate: compare per-rank iteration counts and workload sizes for ranks 1, 17-20 against the faster ranks.
fix: redistribute the computational mesh or data partitions to give ranks 1, 17-20 a smaller share of the work.

Finished analyzing all selected regions.

17. **Path: 1/Karolina/scorep_clover_leaf_64_sum → profile.cubex**
    Time taken: ~5 mins
    1 root; 1,311 call paths, 128 system tree elements

Analyzing call tree...

======================================
REGION 1
----
call_path: "Invisible call tree root/clover_leaf"
severity: 81.6%
slowest_ranks: 29,28,31,27,77
reason: The runtime is concentrated on a few ranks while most ranks have near-zero work, indicating a severe load imbalance.

investigate: compare the number of leaf cells or computational tasks assigned to ranks 29,28,31,27,77 against the idle ranks to confirm uneven workload distribution.
fix: repartition the mesh or leaf cell ownership so that the heavy ranks receive fewer cells and idle ranks receive more, achieving a more even workload.

==
REGION 2
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/clover_module_clover_init_comms_/MPI_Init_thread"
severity: 58.8%
slowest_ranks: [15]
reason: The runtime distribution shows many ranks idle while rank 15 takes much longer, indicating a strong load imbalance.
investigate: compare per-rank workload sizes and iteration counts for rank 15 against the idle ranks to verify the imbalance.
fix: redistribute the computational domain or data partitions from rank 15 to other ranks to achieve a more even workload.

==
REGION 3
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/clover_module_clover_init_comms_/acc_init"
severity: 57%
slowest_ranks: 16
reason: The distribution shows a handful of ranks taking ~5 s while many ranks report near-zero time, indicating a strong load imbalance.
investigate: compare per-rank iteration counts and communication workload for rank 16 versus the idle ranks to verify the imbalance.
fix: redistribute the initialization work (e.g., mesh partitions or communication buffers) from the overloaded rank to the idle ranks to balance the runtime.

==
REGION 4
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/initialise_/start_/build_field_"
severity: 52.6%
slowest_ranks: 89
reason: Many ranks have zero runtime while a subset run about 7–8 s, indicating a strong load imbalance.
investigate: compare per-rank iteration counts or assigned mesh elements between the idle ranks (0 s) and rank 89 to verify unequal workload distribution.

fix: rebalance the domain decomposition to give each MPI rank a more even share of the mesh partitions.

====================================

REGION 5

----

call_path: "Invisible call tree root/clover_leaf/KERNELS/COMPUTE IDLE"
severity: 18.7%
slowest_ranks: [7,3,2,1,0]
reason: The runtimes form two clusters, with a few ranks taking about 10 s while the majority stay near 7.5–9.3 s, indicating significant load imbalance.
investigate: compare per-rank iteration counts and computational work assigned to the ranks listed as slowest to determine why they execute substantially more work.
fix: redistribute mesh or workload so that the high-runtime ranks receive fewer elements, balancing the computational load across all MPI ranks.

Finished analyzing all selected regions.

18. **Path: 1/Karolina/scorep_clover_leaf_128_sum → profile.cubex**
    Time taken:
    1 root; 1,311 call paths, 256 system tree elements

LLM throws a couldn’t handle request error even though it has analyzed other files that are much larger.

Network error: Error transferring https://api.helmholtz-blablador.fz-juelich.de/v1/chat/completions - server replied: Proxy Error

Server response: <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>502 Proxy Error</title>
</head><body>
<h1>Proxy Error</h1>
<p>The proxy server received an invalid
response from an upstream server.
The proxy server could not handle the request<p>Reason: Error reading from remote server</p></p>
</body></html>

19. Path: 1/Karolina/scorep_clover_leaf_256_sum → profile.cubex
Time taken: ~3 mins
1 root; 1,311 call paths, 512 system tree elements

Analyzing call tree...

======================================
REGION 1
----
call_path: "Invisible call tree root/clover_leaf"
severity: 84%
slowest_ranks: [695, 696, 697, 698, 699, 700, 701, 702]
reason: most ranks execute almost no work while a few ranks spend an order of magnitude more time, indicating severe load imbalance.
investigate: compare per-rank execution times and the amount of mesh/particle data processed by ranks 695-702 against the other ranks.
fix: repartition the computational mesh so that the heavy regions handled by ranks 695-702 are split and distributed more evenly across all MPI ranks.

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REGION 2
----
call_path: "Invisible call tree root/clover_leaf/MAIN_/clover_module_clover_init_comms_/MPI_Init_thread"
severity: 54.1%
slowest_ranks: 346
reason: Half of the MPI ranks report zero runtime while the others spend ~7 s, showing a clear load imbalance.
investigate: compare per-rank iteration counts and assigned mesh/particle workload between the idle ranks (runtime 0) and the slowest rank 346 using the profiling data.
fix: redistribute the mesh partitions so that all ranks receive a comparable portion of the computational work, eliminating the large number of idle ranks.

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REGION 3
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call_path: "Invisible call tree root/clover_leaf/MAIN_/hydro_/TIMESTEP/timestep_module_timestep_/clover_module_clover_min_/MPI_Allreduce"
severity: 88.4%
slowest_ranks: [346]
reason: The runtime distribution shows many ranks idle (0 s) while a few ranks take around 10–11 s, indicating severe load imbalance.
investigate: Compare per-rank iteration counts and workload sizes for rank 346 versus the idle ranks to verify the imbalance.

fix: Redistribute the computational domain so that the work assigned to rank 346 is divided among neighboring ranks.

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REGION 4
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call_path: "Invisible call tree root/clover_leaf/KERNELS/COMPUTE IDLE"
severity: 18.3%
slowest_ranks: [338]
reason: The runtimes form a bimodal distribution with many ranks completing in ~8–10 s while a subset take ~15 s, indicating a strong load imbalance.
investigate: compare per-rank computational work or iteration counts between the fast (~8–10 s) and the slow (~15 s) ranks using profiling data.
fix: redistribute mesh or particle partitions to balance the workload so that all ranks perform a similar amount of work.

Finished analyzing all selected regions.

CALL TREE WITH 2 ROOTS

20. Path: 1/ hybrid-POP /scorep_profile.1687543 → profile.cubex
Time taken: ~5 mins
2 roots; 712 call paths, 192 system tree elements

Analyzing call tree...

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REGION 1
----
call_path: "Invisible call tree root/main"
severity: 0%
slowest_ranks: []
reason: All ranks report zero time for this call, indicating that its work is fully captured in descendant calls rather than at this node.
investigate: compare the per-rank runtimes of the child calls beneath this node to verify that work is evenly distributed across ranks.
fix: no balancing needed at this level; ensure the workloads of the child functions are distributed more evenly if they exhibit imbalance.

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REGION 2
----
call_path: "Invisible call tree root/main/MPI_Init_thread"
severity: 95.33%

slowest_ranks: [32]
reason: Most ranks have near-zero runtime while a few ranks take several seconds, indicating a severe load imbalance during MPI initialization.
investigate: Compare the MPI_Init_thread execution time and thread count on each rank to see why rank 32 is slower.
fix: Align the thread support level or thread count across all ranks (e.g., request the same number of threads or use single-thread mode) to balance the initialization cost.

==
REGION 3
----
call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/!$omp parallel @xshells_big.cpp:1578/RadSph"
severity: 71.0%
slowest_ranks: [16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176]
reason: A handful of ranks take more than five seconds while the majority stay near two seconds, indicating a strong load imbalance.
investigate: compare per-rank iteration counts and computational work for the listed ranks against the average ranks to confirm the imbalance.
fix: redistribute the domain or mesh partitions so that the heavy ranks receive less work, balancing runtimes across all ranks.

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REGION 4
----
call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/!$omp parallel @xshells_big.cpp:1578/!$omp implicit barrier @xshells_big.cpp:1614"
severity: 60.5%
slowest_ranks: [48, 16, 32, 96, 64]
reason: A few MPI ranks take more than twice the average time, indicating a serious load-imbalance.
investigate: compare per-rank iteration counts and mesh-partition sizes for the listed ranks against the typical ranks.
fix: repartition the mesh so that the computational load is distributed more evenly across all MPI ranks.

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REGION 5
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call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/SHqst_to_spat/SHqst_to_spat_mic2_l/!$omp parallel @SHqst_to_spat_mic.c:491"
severity: 58.27%
slowest_ranks: 48

reason: A few MPI ranks experience runtimes more than double the average while the majority cluster tightly around the mean, indicating a clear load imbalance.
investigate: compare per-rank iteration counts and assigned work sizes for rank 48 (and other high-runtime ranks) against the typical ranks to confirm that they are handling disproportionately larger workloads.
fix: redistribute mesh partitions or particle ownership from the overloaded ranks (e.g., rank 48, 16, 32) to the underutilized ranks to equalize per-rank runtimes.

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REGION 6
----
call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/SHqst_to_spat/SHqst_to_spat_mic2_l/!$omp parallel @SHqst_to_spat_mic.c:491/sy32_l"
severity: 53.2%
slowest_ranks: 48
reason: The runtime distribution shows a few ranks (e.g., rank 48) taking more than double the average time while most ranks are clustered around 3–5 s, indicating a clear load imbalance.
investigate: Compare the number of iterations, mesh elements, or particle counts assigned to rank 48 with those of typical ranks to verify why its work is larger.
fix: Repartition the mesh or redistribute the computational workload so that rank 48 receives a smaller share, balancing the load across all ranks.

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REGION 7
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call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/!$omp parallel @xshells_big.cpp:1646"
severity: 51.6%
slowest_ranks: [16, 32, 48, 64, 80, 96, 112, 128, 144, 160]
reason: A small subset of ranks take roughly twice the average time while the majority cluster around 4–5 s, indicating a clear load imbalance.
investigate: compare per-rank iteration counts and workload sizes for ranks 16, 32, 48, 64, 80, 96, 112, 128, 144, and 160 to determine why they incur much higher runtimes.
fix: redistribute the mesh or computational partitions so that the heavy work on those ranks is divided more evenly across neighboring MPI ranks.

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REGION 8
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call_path: "Invisible call tree root/main/step_UBT/explicit_terms_omp_inshell/spat_to_SHqst/spat_to_SHqst_mic2_l/!$omp parallel @spat_to_SHqst_mic.c:67/an32_l"
severity: 49.3%

slowest_ranks: [48]
reason: A small subset of ranks (e.g., rank 48) take more than twice the average time, indicating a clear load-imbalance.
investigate: compare the per-rank iteration counts and computational workload assigned to rank 48 versus the other ranks.
fix: redistribute the mesh or data partitions so that the work carried by rank 48 is shared with surrounding ranks.

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REGION 9
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call_path: "Invisible call tree root/main/step_UBT/!$omp parallel @xshells_big.cpp:2182/solve_state/solve_dn/!$omp barrier @xshells_linop.cpp:964"
severity: 46.1%
slowest_ranks: 48
reason: A few ranks (e.g., rank 48) take about double the time of the majority, showing a clear load imbalance.
investigate: compare per-rank iteration counts and workload sizes for rank 48 against the typical ranks to confirm the excess computational work.
fix: redistribute the mesh or domain partitions so that rank 48 receives a smaller share of the total work.

Finished analyzing all selected regions.

21. **Path: 1/ hybrid-POP /scorep_profile.2451145 → profile.cubex**
    Time taken:
    2 roots; 894 call paths, 192 system tree elements

LLM throws a couldn't handle request error even though it has analyzed other files that are much larger.

Analyzing call tree...

====================================
REGION 1
----
call_path: "Invisible call tree root/main"
severity: 73.98%
slowest_ranks: [8,16,24,32,48,56,64,72,80,88,96,104,112,120,128,136,144,152,160,168,176,184]
reason: The runtime distribution shows a small subset of ranks taking roughly ten times longer than the average, while the majority finish in 1–3 s, indicating severe load imbalance.
investigate: compare the number of computational elements (e.g., mesh cells or particles) assigned to each of the listed ranks and their communication volume to verify why they incur far higher runtimes.
fix: repartition the domain so that the heavy computational chunks owned by the slow ranks are evenly redistributed across all MPI ranks.

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REGION 2
----
call_path: "Invisible call tree root/main/MPI_Init_thread"
severity: 74%
slowest_ranks:
0,8,16,24,32,40,48,56,64,72,80,88,96,104,112,120,128,136,144,152,160,168,176,184,192,200,208,216,224,232,240,248
reason: A small subset of ranks repeatedly take ~10-12 s while the majority finish in ~1-2 s, indicating severe load imbalance.
investigate: compare the per-rank execution times for the listed high-runtime ranks against the neighboring low-runtime ranks to confirm the disparity.
fix: redistribute the work assigned to the high-runtime ranks (e.g., rebalance mesh or particle partitions) so that each MPI rank receives a comparable amount of computation.

=====================================
REGION 3
----
call_path: "Invisible call tree root/main/init_sh/MPI_Barrier"
severity: 74.0%
slowest_ranks:
[8,16,24,32,40,48,56,64,72,80,88,96,104,112,120,128,136,144,152,160,168,176,184]
reason: The runtime distribution shows a few ranks spending an order of magnitude more time than the majority, indicating load imbalance due to disproportionately large workload on those ranks.
investigate: compare per-rank iteration counts and workload sizes for the listed ranks to verify they receive significantly more work than the others.
fix: redistribute the large data partitions or computation blocks away from the identified ranks to achieve a more even workload across MPI ranks.

Network error: Error transferring https://api.helmholtz-blablador.fz-juelich.de/v1/chat/completions - server replied: Proxy Error
Server response: <!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>502 Proxy Error</title>
</head><body>
<h1>Proxy Error</h1>
<p>The proxy server received an invalid
response from an upstream server.
The proxy server could not handle the request<p>Reason: Error reading from remote server</p></p>
</body></html>